

Supplementary Material: Memory Access Windows

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This supplement gives the continuous formulation used to prove the temporal-isolation guarantee stated in Section IV of the paper. It is not part of the six-page technical body.

APPENDIX

The main text reasons about the discrete schedule σ . For the proofs we introduce a continuous formulation as a helper. For a flow $f = (i, j)$ (tile i to home slice j), the schedule induces the piecewise-constant *access function*

$$A_f(t) = \sigma_i^j[\lfloor t/\tau \rfloor \bmod L] \in \{0, 1\}, \quad (1)$$

which is 1 exactly during f 's enabled slots. Let $o_f^l(t) \in \{0, 1\}$ indicate that a packet of an in-flight transaction of flow f occupies shared resource l at time t , and let P_f be the (fixed) set of resources f uses. The interference of g on f over $[0, T]$ is

$$I_{fg} = \int_0^T \sum_{l \in P_f \cap P_g} o_f^l(t) o_g^l(t) dt. \quad (2)$$

By definition of the drain time δ , every packet present at time t belongs to a transaction injected at some $s \in [t - \delta, t]$ with $A_f(s) = 1$. Define the *dilated* access function

$$\hat{A}_f(t) = \max_{s \in [t - \delta, t]} A_f(s), \quad (3)$$

i.e. each enabled window extended forward by δ . Then

$$o_f^l(t) = 1 \implies \hat{A}_f(t) = 1. \quad (4)$$

Lemma 1. *If $\hat{A}_f(t) \hat{A}_g(t) = 0$ for all t , then $I_{fg} = 0$.*

Proof. Fix $l \in P_f \cap P_g$ and t . If $o_f^l(t) = o_g^l(t) = 1$ then by (4) $\hat{A}_f(t) = \hat{A}_g(t) = 1$, contradicting the hypothesis. Hence $o_f^l(t) o_g^l(t) = 0$ for every l and every t (all but a set of isolated instants of total duration zero), so the integrand in (2) vanishes and $I_{fg} = 0$. $\square$

Proposition 1. *Suppose that on every shared resource $l \in P_f \cap P_g$ the schedule enables f and g in disjoint slot sets separated by at least $g \geq \lceil \delta/\tau \rceil$ idle ($\sigma = 0$) slots. Then $\hat{A}_f \hat{A}_g \equiv 0$, and therefore $I_{fg} = 0$: the two flows are temporally isolated.*

Proof. Consider any enabled window of f that ends at a slot boundary t_0 . By (3), $\hat{A}_f$ is 1 on that window and remains 1 for at most δ afterward, i.e. up to $t_0 + \delta$. The hypothesis places at least g idle slots – duration $g\tau \geq \lceil \delta/\tau \rceil \tau \geq \delta$ – before the next enabled slot of g , which therefore starts no earlier than $t_0 + \delta$. Hence at no time are both $\hat{A}_f$ and $\hat{A}_g$ equal to 1. The

same holds symmetrically for windows of g preceding f . Thus $\hat{A}_f \hat{A}_g \equiv 0$, and Lemma 1 gives $I_{fg} = 0$. $\square$

Since a contention covert channel between f and g requires $I_{fg} > 0$ (nonzero simultaneous occupancy of a shared resource), Proposition 1 also drives its capacity to zero. Without a guard band the residual overlap per window boundary is at most δ , bounding the leftover capacity.